Years 7–8 standard elaborations — Australian Curriculum v9.0: Technologies

ACiQ | v9.0

Purpose

The standards elaborations (SEs) support teachers to connect curriculum to evidence in assessment so that students are assessed on what they have had the opportunity to learn. The SEs can be used to:

- make consistent and comparable judgments, on a five-point scale, about the evidence of learning in a folio of student work across a year/band
- develop task-specific standards (or marking guides) for individual assessment tasks
- quality assure planning documents to ensure coverage of the achievement standard across a year/band.

Structure

The SEs have been developed using the Australian Curriculum achievement standard. The achievement standard for Technologies describes what students are expected to know and be able to do at the end of each year. Teachers use the SEs during and at the end of a teaching period to make on-balance judgments about the qualities in student work that demonstrate the depth and breadth of their learning.

In Queensland, the achievement standard represents the C standard — a sound level of knowledge and understanding of the content, and application of skills. The SEs are presented in a matrix where the discernible differences and/or degrees of quality between each performance level are highlighted. Teachers match these discernible differences and/or degrees of quality to characteristics of student work to make judgments across a five-point scale.

In Years 7–8 the Learning area achievement standard may be used to assess within and across the Technologies subjects.

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Years 7–8 Australian Curriculum: Technologies achievement standard
By the end of Year 8 students explain how people design, innovate and produce products, services and environments for preferred futures. For each of the 4 prescribed technologies contexts students explain how the features of technologies impact on design decisions, and create designed solutions based on analysis of needs or opportunities. They acquire, interpret and model with spreadsheets and represent data with integers and binary. Students design and trace algorithms; and implement them in a general-purpose programming language. Students create and adapt design ideas, processes and solutions, and justify their decisions against developed design criteria that include sustainability. They communicate design ideas and solutions to audiences using technical terms and graphical representation techniques, including using digital tools. They select appropriate hardware for particular tasks, explain how data is transmitted and secured in networks, and identify cyber security threats. They use a range of digital tools to individually and collaboratively document and manage production processes to safely and responsibly produce designed or digital solutions for the intended purpose. Students manage their digital footprint.
Source: Australian Curriculum, Assessment and Reporting Authority (ACARA), *Australian Curriculum Version 9.0 Technologies for Foundation–10* https://v9.australiancurriculum.edu.au/f-10-curriculum/learning-areas/design-and-technologies_digital-technologies/year-7?view=quick&detailed-content-descriptions=0&hide-ccp=0&hide-gc=0&side-by-side=1&strands-start-index=0&subjects-start-index=0

Years 7–8 Technologies standard elaborations

		A	B	C	D	E
		The folio of student work contains evidence of the following:				
Knowledge and understanding	**Technologies and society**	discerning explanation of how people design, innovate and produce products, services and environments for preferred futures	detailed explanation of how people design, innovate and produce products, services and environments for preferred futures	explanation of how people design, innovate and produce products, services and environments for preferred futures	description of how people design, innovate and produce products, services and environments for preferred futures	statement/s about how people design, innovate and/or produce designed solutions

		A	B	C	D	E
	Technologies contexts	discerning explanation of how the features of technologies impact on design decisions for each of the 4 prescribed technologies contexts: • Engineering principles and systems • Food and fibre production • Food specialisations • Materials and technologies specialisations	detailed explanation of how the features of technologies impact on design decisions for each of the 4 prescribed technologies contexts: • Engineering principles and systems • Food and fibre production • Food specialisations • Materials and technologies specialisations	explanation of how the features of technologies impact on design decisions for each of the 4 prescribed technologies contexts: • Engineering principles and systems • Food and fibre production • Food specialisations • Materials and technologies specialisations	description of how the features of technologies impact on design decisions for each of the 4 prescribed technologies contexts: • Engineering principles and systems • Food and fibre production • Food specialisations • Materials and technologies specialisations	statement/s about the features of technologies for one or more of the prescribed technologies contexts: • Engineering principles and systems • Food and fibre production • Food specialisations • Materials and technologies specialisations
	Digital systems	proficient selection of appropriate hardware for particular tasks	effective selection of appropriate hardware for particular tasks	selection of appropriate hardware for particular tasks	guided selection of appropriate hardware for particular tasks	directed selection of appropriate hardware for particular tasks
		considered explanation of how data is transmitted and secured in networks	detailed explanation of how data is transmitted and secured in networks	explanation of how data is transmitted and secured in networks	description of how data is transmitted and/or secured in networks	statement/s about data transmission and/or security
	Data representation	reasoned representation of data with integers and binary	effective representation of data with integers and binary	representation of data with integers and binary	partial representation of data with integers and binary	fragmented representation of data with integers and/or binary

		A	B	C	D	E
Processes and production skills	**Acquiring, managing and analysing data**	proficient acquisition, interpretation and modelling with spreadsheets	effective acquisition, interpretation and modelling with spreadsheets	acquisition, interpretation and modelling with spreadsheets	partial acquisition, interpretation and/or modelling with spreadsheets	fragmented acquisition, interpretation and/or modelling with spreadsheets
	Investigating and defining	proficient analysis of needs or opportunities for each of the 4 prescribed technologies contexts: • Engineering principles and systems • Food and fibre production • Food specialisations • Materials and technologies specialisations	effective analysis of needs or opportunities for each of the 4 prescribed technologies contexts: • Engineering principles and systems • Food and fibre production • Food specialisations • Materials and technologies specialisations	analysis of needs or opportunities for each of the 4 prescribed technologies contexts: • Engineering principles and systems • Food and fibre production • Food specialisations • Materials and technologies specialisations	superficial analysis of needs or opportunities for each of the 4 prescribed technologies contexts: • Engineering principles and systems • Food and fibre production • Food specialisations • Materials and technologies specialisations	identification of needs or opportunities for one or more of the 4 prescribed technologies contexts: • Engineering principles and systems • Food and fibre production • Food specialisations • Materials and technologies specialisations
	Generating and designing	proficient design and tracing of algorithms	effective design and tracing of algorithms	design and tracing of algorithms	guided design and/or tracing of algorithms	directed design and/or tracing of algorithms
		proficient creation and adaptation of comprehensive design ideas, processes and solutions based on analysis of needs or opportunities	informed creation and adaptation of effective design ideas, processes and solutions based on analysis of needs or opportunities	creation and adaptation of design ideas, processes and solutions based on analysis of needs or opportunities	partial creation and adaptation of simple design ideas, processes and solutions based on analysis of needs or opportunities	fragmented creation and/or adaptation of basic design ideas, processes and solutions based on needs or opportunities

		A	B	C	D	E
		communication of design ideas and solutions to audiences with: • considered use of technical terms • comprehensive use of graphical representation techniques, including using digital tools	communication of design ideas and solutions to audiences with: • effective use of technical terms • detailed use of graphical representation techniques, including using digital tools	communication of design ideas and solutions to audiences using: • technical terms • graphical representation techniques, including using digital tools	communication of design ideas and solutions to audiences with: • superficial use of technical terms and/or • simple use of graphical representation techniques, including using digital tools	communication of design ideas and solutions to audiences with: • fragmented use of technical terms and/or • basic use of graphical representation techniques, including using digital tools
	Producing and implementing	proficient, safe and responsible production of designed or digital solutions for the intended purpose	effective, safe and responsible production of designed or digital solutions for the intended purpose	safe and responsible production of designed or digital solutions for the intended purpose	safe and responsible production of aspects of designed or digital solutions for the intended purpose	safe and responsible production of aspects of designed or digital solutions for the intended purpose, with direction
		proficient implementation of algorithms in a general-purpose programming language	effective implementation of algorithms in a general-purpose programming language	implementation of algorithms in a general-purpose programming language	partial implementation of algorithms in a general-purpose programming language	directed implementation of algorithms
	Evaluating	discerning justification of their decisions against developed design criteria that include sustainability	plausible justification of their decisions against developed design criteria that include sustainability	justification of their decisions against developed design criteria that include sustainability	partial justification of their decisions against aspects of developed design criteria that include sustainability	statement/s about design decisions
	Collaborating and managing	individual and collaborative: • comprehensive documentation • proficient management of production processes using a range of digital tools	individual and collaborative: • detailed documentation • effective management of production processes using a range of digital tools	individual and collaborative documentation and management of production processes using a range of digital tools	individual and collaborative: • partial documentation • guided management of production processes using digital tools	individual and/or collaborative: • fragmented documentation • directed management of production processes using digital tools

		A	B	C	D	E
	Privacy and security	discerning identification of cyber security threats	informed identification of cyber security threats	identification of cyber security threats	partial identification of cyber security threats	directed identification of cyber security threats
		justified management of their digital footprint.	informed management of their digital footprint.	management of their digital footprint.	management of aspects of their digital footprint.	directed management of their digital footprint.

Key	shading emphasises the qualities that discriminate between the A–E descriptors

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